

Unit 3, Lesson 7: Solving Two-Step, Real-World Problems with Fractions or Decimals – Part 2

Benchmark

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

Learning Target

- ☐ I can solve two-step, real-world problems with decimals or fractions.

3.7.1 Warm-Up: Would You Rather?

Would you rather buy 8 Skittles for 25 cents or 62 Skittles for \$1.49?

1. Explain your choice.
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2. Would your answer change if you could choose what flavor the 8 Skittles were but you could not choose the flavors from the bag?

3.7.2 Collaboration: Error Analysis

1. Stefanie solved the following word problem. Her work is shown.
- a. Evaluate her work and determine where she made mistakes.

Problem	Stefanie’s Work
Janelle has $2\frac{1}{4}$ pans of brownies. Each pan made 8 brownies. She gave $\frac{2}{3}$ of the brownies to her friends. How many brownies did Janelle give away?	Step 1: $2\frac{1}{4} \times 8 = \frac{9}{4} \times 8 = \frac{18}{4}$ Step 2: $\frac{18}{4} \div \frac{2}{3} = \frac{18}{4} \times \frac{3}{2} = 6\frac{3}{4}$ Answer: She gave away $6\frac{3}{4}$ brownies.

- b. Help Stefanie understand and learn from her mistakes. Write a brief note to Stefanie explaining what her mistakes were and how she could learn from those mistakes.

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Start
To
Acquire
Knowledge
Experience
Skills

- c. Read your partner’s note to Stefanie and offer suggestions for how to improve their note.

1. While birdwatching, David saw a cardinal $\frac{1}{2}$ of a yard to the left from where he was standing. A bluebird was $\frac{5}{12}$ of a yard to the right of the cardinal. Crystal is standing $1\frac{1}{3}$ yards to the right of David. How far was the bluebird from where Crystal was standing?
2. One gallon of gasoline in Buffalo, New York costs \$2.29. In Toronto, Canada, one liter of gas is \$0.91. There are 3.8 liters in one gallon. How much greater is the cost of gas in either city?

3.7.4 Your Turn!

1. Martha fills her ice cream bowl with $\frac{3}{4}$ cup of ice cream, $\frac{1}{8}$ cup of candy toppings, and the rest with fruit toppings. If the bowl contains a total of $1\frac{1}{2}$ cups, how much of the bowl is made up of fruit toppings?

2. Jasmine has a ribbon that is 12.6 meters long. Sharon has a ribbon that is 3.8 meters shorter. What is the total length of the two ribbons?

3. Before soccer practice, Hernando puts $4\frac{1}{8}$ cups of sports drink in a water bottle with $1\frac{2}{3}$ cups of ice. After a hot afternoon of practice, he drinks $2\frac{1}{5}$ cups of the sports drink and melted ice. How many cups of this mixture are left in the water bottle?

4. A bag of chocolates weighs 1.384 pounds. If Luis gives each of his 6 friends 0.23 pounds of chocolate, how many pounds of chocolate does Luis have left?

3.7.5 Wrap-Up: Ticket Out the Door

1. Priscilla and Dayton sold candy to raise money for their swimming team. Priscilla sold $1\frac{5}{8}$ times as much as Dayton. If Dayton sold $\frac{1}{5}$ of a box of candy, how many total boxes of candy did they sell?